IUPC & ICPC Problem Category Taxonomy & Star Guide

Empirical Analysis of 343 Problems Across 33 Contests (ICPC Dhaka Preliminary, Regional & National IUPCs 2021–2026)

Rank	Category Name	Star Rating	Count	Share	Strategic Contest Role
#1	**Graph Theory**	★★★★★	52	15.2%	Highest volume topic; 1–3 problems in every contest
#2	**Number Theory & Math**	★★★★★	46	13.4%	Bangladeshi specialty; primes, modular & recurrences
#3	**Dynamic Programming (DP)**	★★★★★	44	12.8%	Primary rank separator; dominates Regional onsite (17%)
#4	**Data Structures**	★★★★★	34	9.9%	Range query engine; Segment Tree, Lazy, Fenwick, DSU
#5	**Greedy & Two Pointers**	★★★★☆	28	8.2%	Mid-contest workhorse; sorting & sliding windows
#6	**Combinatorics & Counting**	★★★★☆	28	8.2%	Permutations, Inclusion-Exclusion & Stars and Bars
#7	**Implementation & Simulation**	★★★★☆	28	8.2%	Fast-solve openers; critical in ICPC Prelims (16.3%)
#8	**String Algorithms**	★★★★☆	25	7.3%	Rolling Hash, KMP, Suffix Automaton, Aho-Corasick
#9	**Computational Geometry**	★★☆☆☆	25	7.3%	High difficulty barrier; Convex Hull, sweep & 3D
#10	**Constructive & Interactive**	★★★☆☆	18	5.2%	Rising topic (2024–2026); bitwise basis & query search
#11	**Game Theory**	★★☆☆☆	15	4.4%	Sprague-Grundy, Nim games & DAG minimax strategy

Category 1: Graph Theory ★★★★★

52 Problems (15.2%) • Core Foundational Pillar

Primary Sub-topics & Techniques:

• Trees & LCA: Lowest Common Ancestor via Binary Lifting, Tree Diameter, Centroid Decomposition, Euler Tour Technique.
• Shortest Paths & Traversals: Dijkstra with multi-dimensional states, 0-1 BFS, Multi-source BFS, Grid graphs.
• Spanning Trees: Kruskal with DSU, Boruvka algorithm, MST on complete graphs with number-theoretic weights (GCD, XOR).
• Biconnectivity & Cycles: Tarjan's Bridge and Cut-vertex detection, Block-Cut Trees, Planar graph face traversals.
• Directed Graphs & Flow: Topological Sort, Strongly Connected Components (Tarjan / Kosaraju), Bipartite Matching (Dinic).

Representative Contest Problems:

• ICPC 2025 Regional - D. Save the Wonderland (Guard coverage with radius R via greedy binary lifting)
• ICPC 2025 Prelim - C. Least Compatible Ancestor (Tree ancestry queries on rooted trees)
• IUPC 2022 RUET - D. Maxxxxximum Spanning Tree (Complete graph with edge weight gcd(i, j))
• IUPC 2026 NSU - E. Min-XOR MST (Complete graph with weight A[i] ⊕ A[j] solved via Trie + Boruvka)

Category 2: Number Theory & Mathematics ★★★★★

46 Problems (13.4%) • Core Foundational Pillar

Primary Sub-topics & Techniques:
• Primes, Sieve & Factorization: Linear Sieve of Eratosthenes, Prime Factorization, Legendre formula, Divisor functions.
• Modular Arithmetic: Modulo Inverses (Fermat's Little Theorem, Extended Euclidean), Modular linear equations, Wilson's Theorem.
• GCD, LCM & Multiplicative Functions: Euclidean algorithm, Coprimality, Euler's Totient $\varphi(n)$, Mobius Inversion.
• Linear Recurrences: Matrix Exponentiation for transitions in $O(k^3 \log N)$, Fibonacci generalizations.
• Algebra & Number Properties: Continued fractions, Gray codes, Parity invariants, Lucas Theorem.

Representative Contest Problems:
• ICPC 2023 Prelim - A. Almost Perfect Prime Factor! (Prime factor sharing across array subsets via Sieve)
• IUPC 2023 BUET - J. Eulers Peculiar Dream (Multiplicative functions and Euler Totient prefix summations)
• IUPC 2025 AUST - G. GCD and LCM in Perfect Sync (Prime factor synchronization under GCD/LCM bounds)
• IUPC 2026 DUET - B. Singularity (High-order linear recurrences evaluated via Matrix Exponentiation)

Category 3: Dynamic Programming (DP) ★★★★★

44 Problems (12.8%) • Core Foundational Pillar

Primary Sub-topics & Techniques:
• Tree DP & Rerooting DP: Subtree state aggregation, In-Out Tree DP / Rerooting technique, Tree Knapsack.
• Classical & Grid DP: Longest Subsequences, 2D Grid paths with obstructions, 0/1 & Bounded Knapsack.
• Probability & Expected Value DP: Expected turns/cost on DAGs, Markov chains, Game-theoretic expected values.
• Bitmask DP: Submask iteration $O(3^n)$, Travelling Salesman / Hamiltonian path states, Broken profile DP.
• Digit DP & Optimized DP: Counting integers in range [L, R] with digit constraints; Segment Tree accelerated DP transitions.

Representative Contest Problems:
• ICPC 2021 Prelim - D. Lazy Squirrel (Tree DP paired with rolling hashes for palindromic tree paths)
• ICPC 2025 Regional - E. Love Marriage (Expected value DP with dynamic probability query updates)
• ICPC 2025 Regional - G. Nonogram (1D Nonogram puzzle validation and state reconstruction via DP)
• IUPC 2026 DUET - F. Ebb and Flow (In-Out Tree Rerooting DP)

Category 4: Data Structures ★★★★★

34 Problems (9.9%) • Core Foundational Pillar

Primary Sub-topics & Techniques:
• Segment Trees & Lazy Propagation: Point/Range updates, Lazy propagation, Sparse/Dynamic Segment Trees, Merge-sort trees.
• Binary Indexed Trees (Fenwick / BIT): Inversion counting, 2D BIT, dynamic prefix sums.
• Disjoint Set Union (DSU): Path compression, Union by rank/size, Rollback DSU, DSU on Tree (Sack).
• Monotonic Data Structures: Monotonic Stack/Queue for Range Maximum Queries (RMQ), Sliding window extremum.
• Advanced Trees & Policy-Based DS: Treap, Splay Tree, Ordered Set (PBDS) for dynamic k-th order statistics.

Representative Contest Problems:
• ICPC 2022 Regional - J. Sort It (Dynamic permutation inversion queries using Treap / Splay Trees)
• IUPC 2023 CoU - B. A Lazy Statement (Range update operations utilizing Lazy Segment Trees)
• IUPC 2025 MU - B. Tree Path Price Queries (Heavy-Light Decomposition (HLD) paired with Segment Trees)
• IUPC 2026 NSU - G. Dynamic Ranklist (Live contest scoreboard simulation with PBDS rank lookups)

Category 5: Greedy Algorithms & Two Pointers ★★★★☆

28 Problems (8.2%) • High Frequency Topic

Primary Sub-topics & Techniques:
• Two Pointers & Sliding Window: Multi-array pointer convergence, optimal range variety, bounded sum windows.
• Sorting & Custom Comparators: Multi-criteria priority sorting, inversion count, exchange arguments.
• Interval Scheduling & Priority Queues: Deadline scheduling, interval coverage, greedy heap-based choice.
• Binary Search on Answer: Monotonic predicate checking combined with greedy verification.

Representative Contest Problems:
• ICPC 2022 Prelim - J. Product Quality Analyst (Prefix sum balance and two pointers optimization)
• ICPC 2024 Regional - C. Cut the Stick, Share You Must (Binary search on answer + greedy validation)
• IUPC 2023 BUET - C. The Very Last Exam (Priority queue-based greedy deadline scheduling)
• IUPC 2025 AUST - A. Max-Min Madness (Two pointers on sorted element differences)

Category 6: Combinatorics & Counting ★★★★☆

28 Problems (8.2%) • High Frequency Topic

Primary Sub-topics & Techniques:
• Permutations, Combinations & Factorials: nCr mod P via precomputed factorials and modular inverse, Catalan numbers.
• Principle of Inclusion-Exclusion (PIE): Overlapping set unions, Derangements, Coprime tuples counting.
• Stars and Bars & Bijections: Allocating identical balls into distinct boxes under boundary constraints.
• Burnside's Lemma & Polya Enumeration: Distinct colorings under rotational/reflectional symmetry.

Representative Contest Problems:
• ICPC 2025 Prelim - F. Over Counting (Combinatorial counting with subtle overlapping set adjustments)
• ICPC 2025 Regional - A. Mission Hexa (Hexagonal grid shortest path combinatorics modulo 10^9+7)
• IUPC 2022 AUST - G. Creative Kindergarten (Complex constraints resolved via Inclusion-Exclusion)
• IUPC 2026 BUET - E. Chorki (Circular rotational equivalence using Burnside's Lemma)

Category 7: Implementation & Simulation ★★★★☆

28 Problems (8.2%) • High Frequency Topic

Primary Sub-topics & Techniques:
• Step-by-Step Simulation: Direct execution of stated rules (cricket scoring, match tiebreakers, penalty rules).
• Grid & Matrix Operations: Coordinate manipulation, 2D matrix rotations, spiral scans, flood fill.
• Date, Time & String Parsing: Calendar leap year conversions, escape sequence parsing (\\, \0), syntax checks.
• Ad-hoc Logic & Case Work: Direct boundary check, small search spaces, exhaustive case verification.

Representative Contest Problems:
• ICPC 2021 Prelim - A. Omicron Juice & B. Omicron Juice Again (Multi-glass liquid transfer simulation)
• ICPC 2021 Regional - E. Calendars (Leap year rule simulation and date difference shifting)
• ICPC 2025 Regional - J. C-Style String Length (Parsing escape sequences \\ and \0 in string literals)
• IUPC 2022 RUET - A. Maiden Over (Cricket over simulation and maiden over counting)

Category 8: String Algorithms ★★★★☆

25 Problems (7.3%) • High Frequency Topic

Primary Sub-topics & Techniques:
• String Matching & Rolling Hash: Double polynomial rolling hashing (mod 10^9+7, 10^9+9), KMP algorithm (π-table), Z-algorithm.
• Advanced Automata & Suffix Structures: Suffix Automaton (SAM), Suffix Array with LCP array, Aho-Corasick Automaton.
• Palindromic Structures: Manacher's Algorithm, Palindromic Tree (Eertree).
• Prefix Trees (Tries): Standard Trie for multi-string retrieval, Binary Trie for bitwise operations.

Representative Contest Problems:
• ICPC 2022 Prelim - E. Cyclic Palindrome (Rolling hash on cyclic string rotations)
• ICPC 2024 Regional - D. CatGPT (Multi-pattern dictionary matching using Aho-Corasick Automaton)
• ICPC 2025 Regional - I. Two Strings Attached (String concatenation equality via Rolling Hash / Z-Algorithm)
• IUPC 2025 UU - H. Substring Symphony (Distinct substring counting via Suffix Automaton)

Category 9: Computational Geometry ★★☆☆☆

25 Problems (7.3%) • Specialized Topic

Primary Sub-topics & Techniques:
• Convex Hull & Polygons: Monotone Chain / Graham Scan, Point-in-polygon tests (ray casting), Polygon center of mass.
• Vectors & Intersections: Cross product, Dot product, Line segment intersections, Half-plane intersections.
• Angular & Sweep Line: Radial angular sweep, Rotating calipers, Segment intersection sweep line.
• Circles & 3D Geometry: Circle-circle intersections, Tangents, Spherical coordinates & Great Circle distances.

Representative Contest Problems:
• ICPC 2021 Regional - D. Der Stern (Great circle distance on 3D spherical coordinates)
• ICPC 2023 Prelim - F. Mesmerizing Fireflies (2D point cluster angular sweep)
• IUPC 2022 RUET - F. Unbalanced Polygon (Centroid of mass calculation and point-in-polygon check)
• IUPC 2023 SUST - D. CONSEPHULL (Dynamic Convex Hull perimeter and area maintenance)

Category 10: Constructive Algorithms & Interactive ★★★☆☆

18 Problems (5.2%) • Regular Topic

Primary Sub-topics & Techniques:
• Bitwise Linear Basis: Gaussian elimination over GF(2), XOR Basis for spanning subsets & reachability.
• Interactive Search: Querying hidden string, graph, or number with restricted query budgets.
• Invariants & Parity: Grid flipping invariants, 2D matrix checkerboard parity, sequence reordering invariants.

Representative Contest Problems:
• ICPC 2021 Regional - F. Flip (Gaussian elimination on linear bit systems)
• ICPC 2022 Regional - F. 2x2 Flip (Invariant analysis on 2x2 binary matrix flips)
• IUPC 2024 IUT - A. Missing String (Interactive secret string recovery via count queries)
• IUPC 2026 BUET - F. Mandatory XOR Problem (Linear basis subset construction under constraints)

Category 11: Game Theory ★★☆☆☆

15 Problems (4.4%) • Specialized Topic

Primary Sub-topics & Techniques:
• Nim & Sprague-Grundy Theorem: Impartial games, XOR sum of Grundy values, Subtraction games, MEX rule.
• Games on Trees & DAGs: Minimax search, Retrograde analysis on game states, Subgame perfect equilibria.
• Turn-based Invariants: Parity control, Symmetry strategies, Misere Nim play.

Representative Contest Problems:
• ICPC 2021 Prelim - G. Alice, Bob and Tree (Minimax turn-based game played on tree nodes)
• ICPC 2023 Regional - G. Pess Chuzzle & K. Strategies in Sequential Games (Chess pawn retrograde analysis)
• IUPC 2022 AUST - D. Nim Heaps (Stone heap game evaluated with Sprague-Grundy theorem)
• IUPC 2024 CUET - D. Max SMEX (Subarray game evaluated using MEX properties)